ANU AGARWAL

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EDUCATION

University of Illinois, Urbana-Champaign Jan 2024 - May 2025
Master's, Computer Science (Focus: Generative AI, Machine Learning, Computer Vision) *CGPA 4.0*

Delhi Technological University, India Aug 2016 - Jun 2020
Bachelor's, Computer Science *CGPA 9.6*

EXPERIENCE

National Center of Supercomputing Applications (NCSA) Sep 2024 – Present
Graduate Research Assistant - Python, Pytorch, Roboflow, sklearn *Champaign, IL*

- Leveraging **YoloV8 + LSTM** to perform real-time activity level detection of pigs using video-feed from the farm.
- Developing a robust, coordinate-based Piglet Behavior Prediction framework using state-of-the-art object tracking and activity recognition techniques.

Goldman Sachs Jul 2021 – Dec 2023
Associate Quantitative Strategist - Systematic Volatility Trading, Java, Python, Slang *Bengaluru, India*

- Developed an algorithm to compute stock volatility values outside market hours, and a Java service around this to use the generated volatility values in real-time pricing of products, **cutting down on Vol-Arb losses** by ∼**52%**
- Developed a Kafka-based service that supplies real-time volatility fit-quality data to the pricing platform, with projected annual savings of ∼**12 million USD** to the firm.
- Collaborated with cross-functional teams to create a **Trading Pattern Recognition** Framework to analyze client trading patterns, flag the arbitrage flow in advance, and adjust spreads accordingly.

Samsung Semiconductors Aug 2020 – Jul 2021
Senior Engineer - Python, C++ *Bengaluru, India*

- Collaborated with Advanced Sensor Algorithms Development and **Research Team**, primarily in C++ & Python
- Developed an end-to-end pipeline for **3D scene reconstruction** from scratch, to generate 3D surface model of any scene from 3D PointClouds and RGB images.
- Enhanced the performance using **OpenMP** multithreading and **OpenCL GPGPU** programming.
- Worked with the wider advanced algorithms team to come up with a novel neural network architecture to solve the Latency-Accuracy trade-off for different ISP tasks.

PROJECTS

Leveraging Contextual Sparsity in Quantized LLMs | *LLM Optimization, Quantization*

- Developed a framework that predicts "contextual sparsity" within quantized LLMs (LLama2) with ∼**90%** accuracy, and significantly reduce their inference time by only loading the non-sparse weights in the memory. Conducted experiments to study the impact of induced sparsity on model size, inference time and perplexity. (April 2024)

Edge-device friendly Diffusion Models for Image Generation | *Deep Learning, Computer Vision*

- Developed a *tiny* diffusion model from scratch, with interspersed convolution and attention blocks in the U-Net architecture, achieving **53% reduction in model size** without impacting the image generation quality.
- Optimized Latent Diffusion Model through Progressive Distillation, **reducing sampling steps by 50%** while maintaining image quality at 24-second generation time (April 2024)

PUBLICATIONS

- *Potent Real-Time Recommendations using Multi-Model Contextual Reinforcement Learning* published in IEEE Transactions on Computational Social Systems (Pages 581-593) (August 2021)
- *Personalized and Dynamic top-k Recommendation System using Context Aware Deep Reinforcement Learning* presented in IEEE COMPSAC, 2021 (Madrid, Spain), published in IEEE 45th Annual COMPSAC (Pages 238-247) (July 2021)

TECHNICAL SKILLS

Languages: C/C++, Java, Python, JavaScript, TypeScript, SQL, Slang, WebPPL
Libraries: PyTorch, Tensorflow, Pandas, NumPy, Scikit-learn, Matplotlib, Diffusers, Transformers, OpenCV
Frameworks: Apache Superset, Apache Kafka, Open3D, Tableau, Hive, Presto, React, Node
Cloud Platforms: Amazon Web Services (AWS), Google Cloud Platform (GCP)
ML/Vision Paradigms: Reinforcement Learning, RLHF, Deep Learning, Natural Language Processing, LLM finetuning, LLM Optimization, Machine Unlearning, Conversational AI, 3D Reconstruction, Computational Photography